25-26年春 19r 小测1

1. $u_{xx}+4u_{xy}\cdot x+(4x^2-1)u_{yy}+2u_y=x^2y \qquad x,y\in R$

$$\begin{cases} u|_{y=x^2}=0 \\ \frac{\partial u}{\partial y}\Big|_{y=x^2}=e^x \end{cases}$$

判断方程类型并求u

$\Delta=1>0$ 双曲型

$$\begin{cases} \xi=y-x^2+x \\ \eta=y-x^2-x \end{cases} \qquad u_{\xi\eta}=\frac{1}{8}(\xi+\eta)$$

$$u=F(\xi)+G(\eta)+\frac{1}{16}\xi\eta(\xi+\eta)$$

$$\begin{cases} u|_{y=x^2}=0 \\ \frac{\partial u}{\partial y}\Big|_{y=x^2}=e^x \end{cases} \Rightarrow \begin{matrix} F(x)+G(-x)=0 \\ F'(x)+G'(-x)=\frac{1}{8}x^2+e^x \end{matrix} \Rightarrow \begin{matrix} F(x)+G(-x)=0 \\ F(x)-G(-x)=\frac{1}{24}x^3+e^x+c \end{matrix}$$

$$F(x)=\frac{1}{48}x^3+\frac{1}{2}e^x+\frac{c}{2}$$

$$\Rightarrow G(x)=\frac{1}{48}x^3-\frac{1}{2}e^{-x}-\frac{c}{2}$$

$$u=\frac{1}{6}(y-x^2)^3+\frac{1}{2}e^{y-x^2+x}-\frac{1}{2}e^{-y+x^2+x}$$

2.

$$\begin{cases} \Delta u=x^2-y^2 \\ u|_{y=x}=2026x \\ u|_{y=-x}=2024x \\ u|_{\Gamma}=2025x+y \end{cases}$$

$D=\{(x,y)\mid x^2+y^2\le 1,\ y\ge|x|\}$

Γ 表示边界

~~求u~~ $v=u-(2025x+y)$

求极坐标下v方程，并求u

$$\begin{cases} \Delta v=r^2\cos 2\theta \\ v|_{\theta=\frac{\pi}{4}}=v|_{\theta=\frac{3}{4}\pi}=0 \\ v|_{r=1}=0 \end{cases}$$

$\phi''+\lambda\phi=0$

$\lambda_n=n^2$

$$\phi_n(\theta)=\begin{cases} A_n(\cos n\theta-\sin n\theta) & n\text{奇数} \\ A_n\cos n\theta & n\equiv 2\pmod 4 \\ B_n\sin n\theta & n\equiv 0\pmod 4 \end{cases}$$

$$\begin{cases} r^2R_2''+rR_2'-4R_2=r^4 \\ R_2(1)=0 \end{cases} \qquad n\ne 2 \begin{cases} r^2R_n''+rR_n'-n^2R_n=0 \\ R_n(1)=0 \\ R_n(r)=0 \end{cases}$$

$$R_2(r)=\frac{r^4-r^2}{12}$$

$$v=\frac{r^4-r^2}{12}\cos 2\theta$$

$$u=2025x+y+\frac{(x^2-y^2)}{12}(x^2+y^2-1)$$